Using Donkey Car And ORB SLAM 3 to Map Racing Tracks 2/26/2026 Status

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Donkey Car-Based SLAM

- Donkey Car is an affordable platform for exploring autonomous vehicles
- Use ORB SLAM 3 to map EBU2 track
- Provide new environments for Donkey Sim to train cars
- Contribute back to UCSD and Donkey Car open source community

Read ORB-SLAM3 Paper

ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial and Multi-Map SLAM

Carlos Campos*, Richard Elvira*, Juan J. Gómez Rodríguez, José M.M. Montiel and Juan D. Tardós

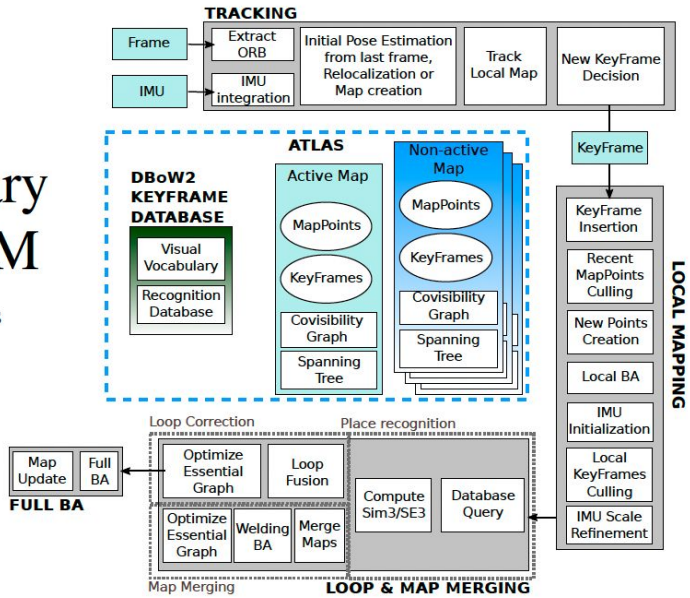

Figure 1: Main system components of ORB-SLAM3.

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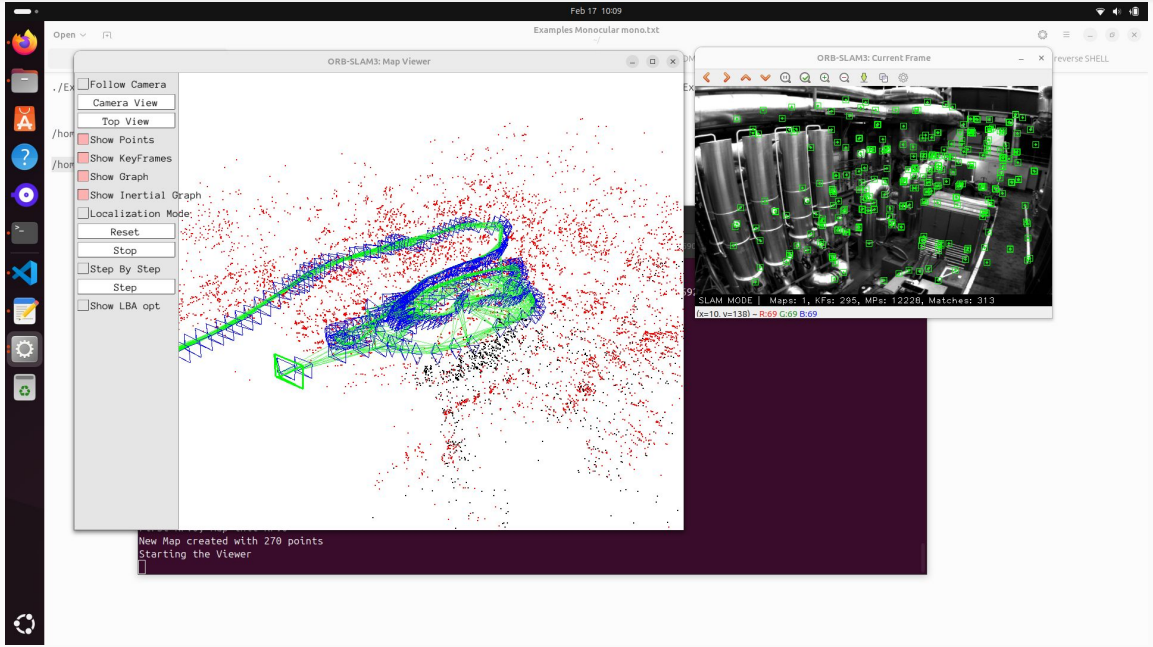
Ran ORB-SLAM3 Example



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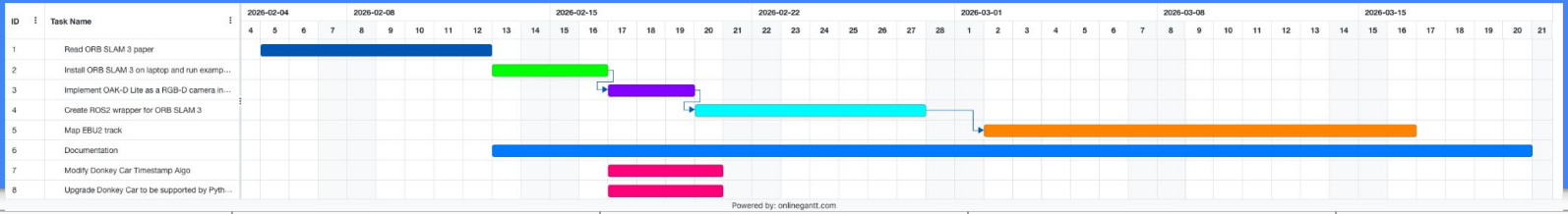
Timeline



Task	Start Date	End Date	Dependencies	Completion %
Read ORB SLAM 3 paper	2/5/2026	2/12/2026	None	100
Install ORB SLAM 3 on laptop and run examples	2/13/2026	2/16/2026	None	100
Implement OAK-D Lite as a RGB-D camera in ORB SLAM 3	2/17/2026	2/19/2026	ORB SLAM 3 running on laptop	50
Modify Donkey Car Timestamp Algo	**2/17/2026**	**2/20/2026**	**None**	0
Upgrade Donkey Car to be supported by Python 3.12			***None***	0
~~Create ROS2 wrapper for ORB SLAM 3~~	~~2/20/2026~~	~~2/27/2026~~	~~Subtask to implement SLAM~~	~~0~~
Map EBU2 track	3/1/2026	3/14/2026	Implement SLAM	0
Documentation	2/13/2026	3/21/2026	Document as we go	0

Reference

- ORB-SLAM3 Paper
- ORB-SLAM3 Github